



SYDNEY GIRLS HIGH SCHOOL

2008

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Diagrams are NOT drawn to scale.
- All necessary working should be shown in every question.
- Start each question on a new page.

Total marks – 84

- Attempt Questions 1 – 7.
- All questions are of equal value.

This is a trial paper ONLY. It does not necessarily reflect the format or contents of the 2008 HSC Examination in this subject.

SGHS Extension 1 THSC 2008

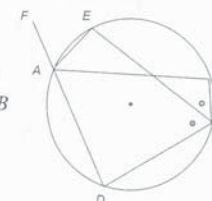
Question 1:

- Solve for x if $\frac{3x}{x-1} < 2$ [3]
- Find the exact value of $\int_0^1 xe^{-x^2} dx$ [2]
- Find $\frac{d}{dx}(\sin^{-1} \sqrt{x})$ [2]
- If A is the point (3,-2) and B is (-3,1) find the point P that divides AB externally in the ratio 2:1. [2]
- If the limiting sum of a geometric progression is 5 and the first term is 3, find the common ratio. [3]

Question 2:

- Sketch the curve $y = 3 \sin^{-1} 2x$ [2]
- If $2 \tan \theta = 3$ and θ is acute, find the exact value of $\frac{3 \sin \theta + 2 \cos \theta}{\sec \theta - \operatorname{cosec} \theta}$ [3]
- The roots of $x^3 - 3x^2 + 6x - 5 = 0$ are α, β and γ . Find the value of
 - $(\alpha+1) + (\beta+1) + (\gamma+1)$ [2]
 - $(\alpha+1)(\beta+1)(\gamma+1)$ [2]

- In the diagram, DA is extended to F and EC bisects $\angle BCD$. Prove that AE bisects $\angle FAB$ [3]



Question 3:

- a) Find the constant term in the expansion $\left(\frac{2}{x} - \frac{x^2}{3}\right)^{12}$ [3]
- b) Find the value of k if $\int_0^k \sqrt{1+x} \, dx = \frac{14}{3}$ [3]
- c) Given that $x = 2$ is an approximation to $x^3 - 9 = 0$, use one application of Newton's Method to find a better approximation. [3]
- d) Use induction to show that: $1 \times 3 \times 5 \times \dots \times (2n-1) = \frac{(2n)!}{2^n \cdot n!}$ [3]

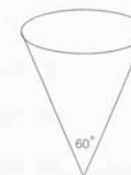
Question 4:

- a) Show that $7^n + 2$ is divisible by three for all positive integral n . [3]
- b) Consider the polynomial $P(x) = x^3 - (k+1)x^2 + kx + 12$
- i) Find the remainder when $P(x)$ is divided by $A(x) = x - 3$ [2]
- ii) Find k if $P(x)$ is divisible by $A(x)$ [2]
- iii) For this value of k , find the zeros of $P(x)$ [2]
- c) Find $\int \frac{x \, dx}{(25 + x^2)^{3/2}}$ using the substitution $u = 25 + x^2$ [3]

Question 5:

- a) A projectile is fired at an angle of $\tan^{-1}\left(\frac{5}{12}\right)$ to the horizontal with an initial velocity of 130 ms^{-1} . Using $g = 10 \text{ ms}^{-2}$,
- i) Derive the equations for the horizontal and vertical position of the projectile at time t . [2]
- ii) Determine the horizontal range of the projectile. [3]

- b) A right circular cone with vertical angle 60° is being filled with a liquid. The liquid is being poured into the cone at a rate of $3\pi \text{ cm}^3 \text{ s}^{-1}$.



- i) Show that the volume of the cone is given by $V = \frac{\pi h^3}{9}$, where h is the height of the liquid [1]

- ii) Find the rate the height of the liquid is increasing when that height is 9 cm . [4]

- c) Find a general solution to $\cos 2x = \sin x$ [2]

Question 6:

- a) A particle is moving in a straight line with an initial velocity of 48ms^{-1} .
At any time t , the acceleration in ms^{-2} is given by $\ddot{x} = -6t$. Find
- The distance from its initial position when it stops [2]
 - The velocity as which it is travelling when it returns to its initial position. [2]
- b) A particle moving on a horizontal line has a velocity $v(\text{ms}^{-1})$, given by $v^2 = 64 + 24x - 4x^2$
- Prove the particle moves in simple harmonic motion [1]
 - Find the centre of motion [1]
 - Write down the period and amplitude [2]
 - If initially, the particle is at the centre of motion and is moving to the left, write down an expression for the displacement as a function of time. [1]
- c) i) Write the expression $\sqrt{3}\cos\theta + \sin\theta$ in the form $R\cos(\theta - \alpha)$ [1]
- Hence or otherwise solve $\sqrt{3}\cos\theta + \sin\theta = 1$, for $0^\circ \leq \theta \leq 360^\circ$. [2]

Question 7:

- a) The maximum charge of a battery that requires recharging is 12 volts. After 1 minute on the recharger, the charge is measured as 6 volts and after a further minute the charge has risen to 8 volts. Assuming the rate of increase of the charge is given by $\frac{dQ}{dt} = k(12 - Q)$,
- Show this equivalent to $Q = 12 - Ae^{-kt}$ [2]
 - Find the value of the charge on the battery at the time it is placed on the recharger. [2]
 - Find the exact value of the battery after it has been on the recharger for 3 minutes. [3]
- b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. If PQ passes through the point $(4a, 0)$
- Show that $pq = 2(p + q)$ [2]
 - Find the locus of M, the midpoint of PQ. [3]

Question 1:

- a) Solve for x if $\frac{3x}{x-1} < 2$

[3]

Solution:

$$\frac{3x}{x-1} < 2 \Rightarrow \frac{3x}{x-1} - 2 < 0 \Rightarrow \frac{3x - 2(x-1)}{x-1} < 0$$

$$\Rightarrow \frac{x+2}{x-1} < 0$$

$$\text{or } 2(x-1)^2 > 3x(x-1)$$

$$\Rightarrow 2(x-1)^2 - 3x(x-1) > 0$$

$$\Rightarrow (x-1)(2x-2-3x) > 0$$

$$\Rightarrow (x-1)(-x-2) > 0$$

$$\text{or } (x-1)(x+2) < 0$$

$$\therefore -2 < x < 1$$

- b) Find the exact value of $\int_0^1 x e^{-x^2} dx$

[2]

Solution:

$$\int_0^1 x e^{-x^2} dx = -\frac{1}{2} \int_0^1 -2x e^{-x^2} dx$$

$$= -\frac{1}{2} [e^{-x^2}]_0^1$$

$$= \frac{1}{2} [e^{-x^2}]_1^0$$

$$= \frac{1}{2} [1 - e^{-1}] = \frac{1}{2} \left[1 - \frac{1}{e} \right]$$

- c) Find $\frac{d}{dx} (\sin^{-1} \sqrt{x})$

[2]

Solution:

$$\frac{d}{dx} (\sin^{-1} \sqrt{x}) = \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{1-x}}$$

$$= \frac{1}{2\sqrt{x(1-x)}}$$

- d) If A is the point (3,-2) and B is (-3,1) find the point P that divides AB externally in the ratio 2:1.

[2]

Solution:

$$\begin{aligned} P(x, y) &= \left(\frac{k_1 x_2 + k_2 x_1}{k_1 + k_2}, \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2} \right) \\ &= \left(\frac{2(-3) - 1(3)}{2-1}, \frac{2(1) - 1(-2)}{2-1} \right) \\ &= \left(\frac{-6-3}{1}, \frac{2+2}{1} \right) \\ &= (-9, 4) \end{aligned}$$

- e) If the limiting sum of a geometric progression is 5 and the first term is 3, find the common ratio.

[3]

Solution:

$$S_{\infty} = \frac{a}{1-r}$$

$$5 = \frac{3}{1-r}$$

$$5 - 5r = 3$$

$$2 = 5r$$

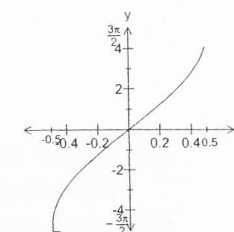
$$\therefore r = \frac{2}{5}$$

Question 2:

- a) Sketch the curve $y = 3 \sin^{-1} 2x$

[2]

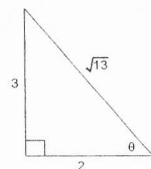
Solution:



- b) If $2 \tan \theta = 3$ and θ is acute, find the exact value of $\frac{3 \sin \theta + 2 \cos \theta}{\sec \theta - \operatorname{cosec} \theta}$

Solution:

$$\begin{aligned} \frac{3 \sin \theta + 2 \cos \theta}{\sec \theta - \operatorname{cosec} \theta} &= \frac{3\left(\frac{3}{\sqrt{13}}\right) + 2\left(\frac{2}{\sqrt{13}}\right)}{\frac{1}{\frac{2}{3}} - \frac{1}{\frac{3}{2}}} \\ &= \frac{\frac{9}{\sqrt{13}} + \frac{4}{\sqrt{13}}}{\frac{3}{2} - \frac{2}{3}} \\ &= \frac{\frac{13}{\sqrt{13}}}{\frac{5}{6}} = 6 \end{aligned}$$



[3]

- c) The roots of $x^3 - 3x^2 + 6x - 5 = 0$ are α, β and γ . Find the value of

i) $(\alpha+1) + (\beta+1) + (\gamma+1)$

[2]

Solution: $\alpha + \beta + \gamma = -\frac{b}{a} = 3$

$$\therefore (\alpha+1) + (\beta+1) + (\gamma+1) = 3+3 = 6$$

ii) $(\alpha+1)(\beta+1)(\gamma+1)$

[2]

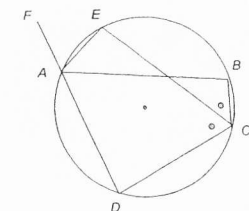
Solution:

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 6$$

$$\alpha\beta\gamma = -\frac{d}{a} = 5$$

$$\begin{aligned} (\alpha+1)(\beta+1)(\gamma+1) &= \alpha\beta\gamma + \alpha\beta + \beta\gamma + \gamma\alpha + \alpha + \beta + \gamma + 1 \\ &= 5 + 6 + 3 + 1 = 15 \end{aligned}$$

- e) In the diagram, DA is extended to F and EC bisects $\angle BCD$. Prove that AE bisects $\angle FAB$



[3]

Solution:

$$\angle ECD = \angle EAF \text{ (exterior angle of cyclic quad)}$$

$$\angle ECB = \angle EAB \text{ (angles on the same arc)}$$

$$\text{Since } \angle ECB = \angle ECD \text{ (given), } \angle EAB = \angle EAF$$

$$\therefore \text{AE bisects } \angle FAB$$

Question 3:

- a) Find the constant term in the expansion $\left(\frac{2}{x} - \frac{x^2}{3}\right)^{12}$

[3]

Solution:

$$U_{r+1} = {}^nC_r a^{n-r} b^r = {}^{12}C_r \left(\frac{2}{x}\right)^{12-r} \left(-\frac{x^2}{3}\right)^r$$

$$\text{The } x \text{ coefficient is } x^{r-12} x^{2r} = x^{3r-12}, \text{ so } x = 4$$

$$U_5 = {}^{12}C_4 \left(\frac{2}{x}\right)^8 \left(-\frac{x^2}{3}\right)^4 = \frac{495 \times 256}{81} = \frac{14080}{9}$$

- b) Find the value of k if $\int_0^k \sqrt{1+x} dx = \frac{14}{3}$

[3]

Solution:

$$\left[\frac{2(1+x)^{\frac{3}{2}}}{3}\right]_0^k = \frac{14}{3}$$

$$\therefore \left[(1+x)^{\frac{3}{2}}\right]_0^k = 7$$

$$\therefore (1+k)^{\frac{3}{2}} - 1 = 7$$

$$(1+k)^{\frac{3}{2}} = 8$$

$$1+k = 4 \rightarrow k = 3$$

- c) Given that $x = 2$ is an approximation to $x^3 - 9 = 0$, use one application of Newton's Method to find a better approximation.

[3]

Solution:

$$\begin{aligned}x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\f(x) &= x^3 - 9 \rightarrow f(2) = 8 - 9 = -1 \\f'(x) &= 3x^2 \rightarrow f'(2) = 12 \\x_1 &= 2 - \frac{f(2)}{f'(2)} = 2 - \frac{-1}{12} = 2\frac{1}{12}\end{aligned}$$

- d) Use induction to show that: $1 \times 3 \times 5 \times \dots \times (2n-1) = \frac{(2n)!}{2^n n!}$

[3]

Solution:

at $n = 1$, $LHS = 1$

$$RHS = \frac{2!}{2^1} = 1 = LHS \rightarrow \therefore \text{true for } n = 1$$

$$\text{assume true for } n = k \rightarrow 1 \times 3 \times 5 \times \dots \times (2k-1) = \frac{(2k)!}{2^k k!}$$

$$\text{prove for } n = k+1, \text{ i.e. prove } 1 \times 3 \times 5 \times \dots \times (2k-1)(2k+1) = \frac{(2k+2)!}{2^{k+1} (k+1)!}$$

$$LHS = 1 \times 3 \times 5 \times \dots \times (2k-1)(2k+1)$$

$$= \frac{(2k)!}{2^k k!} (2k+1) \quad [\text{using assumption}]$$

$$= \frac{(2k)!}{2^k k!} \frac{(2k+1)(2k+2)}{2k+2}$$

$$= \frac{(2k)!}{2^k k!} \frac{(2k+1)(2k+2)}{2(k+1)}$$

$$= \frac{(2k+2)!}{2^{k+1} (k+1)!}$$

$$= RHS$$

$\therefore$ if true for $n = k$, it is true for $n = k+1$.

Since true for $n = 1$, it is true for $n = 2$, then $n = 3$ and so on for all positive integral values of n .

Question 4:

- a) Show that $7^n + 2$ is divisible by three for all positive integral n .

[3]

Solution:

at $n = 1$, $7^1 + 2 = 9 = 3 \times 3$, so is true for $n = 1$

assume true for $n = k \rightarrow 7^k + 2 = 3M$ (M an integer)

prove for $n = k+1$

$$7^{k+1} + 2 = 7 \times 7^k + 2$$

$$= 7 \times (3M - 2) + 2$$

$$= 21M - 14 + 2$$

$$= 21M - 12$$

$$= 3(7M - 4)$$

$\therefore$ if true for $n = k$, it is true for $n = k+1$.

Since true for $n = 1$, it is true for $n = 2$, then $n = 3$ and so on for all positive integral values of n .

- b) Consider the polynomial $P(x) = x^3 - (k+1)x^2 + kx + 12$

- i) Find the remainder when $P(x)$ is divided by $A(x) = x - 3$

[2]

Solution:

$$P(3) = 27 - 9k - 9 + 3k + 12$$

$$= 30 - 6k$$

- ii) Find k if $P(x)$ is divisible by $A(x)$

[2]

Solution:

$$\text{If } P(3) = 0, 30 - 6k = 0$$

$$\text{and } k = 5$$

- iii) For this value of k , find the zeros of $P(x)$

[2]

Solution:

$$P(x) = x^3 - 6x^2 + 5x + 12$$

$$= (x-3)(ax^2 + bx + c) \text{ as } (x-3) \text{ is a factor}$$

$$= (x-3)(x^2 - 3x - 4)$$

$$= (x-3)(x-4)(x+1)$$

$$\therefore x = -1, 3, 4$$

c) Find $\int \frac{x \cdot dx}{(25 + x^2)^{3/2}}$ using the substitution $u = 25 + x^2$

[3]

Solution:

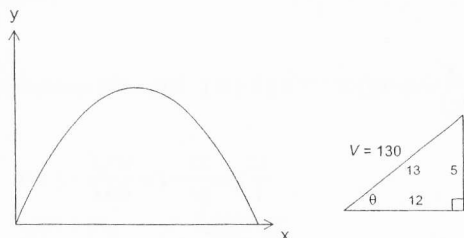
$$\begin{aligned} \text{let } u &= 25 + x^2, \\ \therefore du &= 2x \cdot dx \\ I &= \frac{1}{2} \int \frac{du}{u^{3/2}} = \frac{1}{2} \int u^{-3/2} \cdot du \\ &= \frac{1}{2} \cdot 2u^{-1/2} + c = \frac{1}{\sqrt{u}} + c \\ &= \frac{1}{\sqrt{25 + x^2}} + c \end{aligned}$$

Question 5:

- a) A projectile is fired at an angle of $\tan^{-1}\left(\frac{5}{12}\right)$ to the horizontal with an initial velocity of 130ms^{-1} . Using $g = 10\text{ms}^{-2}$,
- Derive the equations for the horizontal and vertical position of the projectile at time t .
 - Determine the horizontal range of the projectile.

[2]

[3]



Solution:

$$\begin{aligned} \ddot{x} &= 0, & \ddot{y} &= -10 \\ \dot{x} &= c_1, & \dot{y} &= -10t + c_2 \end{aligned}$$

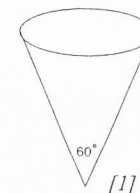
$$\begin{aligned} \text{at } t = 0, \dot{x} &= V \cos \theta, & \dot{y} &= V \sin \theta \\ &= 130 \left(\frac{12}{13} \right), & &= 130 \left(\frac{5}{13} \right) \\ &= 120 = c_1, & &= 50 = c_2 \end{aligned}$$

$$\begin{aligned} \therefore \dot{x} &= 120, & \dot{y} &= -10t + 50 \\ \therefore x &= 120t + c_3, & y &= -5t^2 + 50t + c_4 \\ \text{at } t = 0, x &= 0 = c_3, & y &= 0 = c_4 \\ \therefore x &= 120t, & y &= -5t^2 + 50t \end{aligned}$$

- ii) at $y = 0$, $t = 0, 10$
range at $t = 10$
sub into $x = 120t$ and range is 600 metres

- b) A right circular cone with vertical angle 60° is being filled with a liquid. The liquid is being poured into the cone at a rate of $3\pi\text{cm}^3 \cdot \text{s}^{-1}$.

- i) Show that the volume of the cone is given by $V = \frac{\pi h^3}{9}$, where h is the height of the liquid



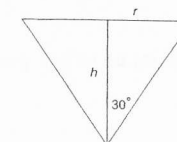
[1]

- ii) Find the rate the height of the liquid is increasing when that height is 9 cm.

[4]

Solution:

$$\begin{aligned} \text{i) } V &= \frac{1}{3} \pi r^2 h & \tan 30^\circ &= \frac{r}{h} \rightarrow \frac{1}{\sqrt{3}} = \frac{r}{h} \text{ and } r = \frac{h}{\sqrt{3}} \\ \therefore V &= \frac{\pi}{3} \left(\frac{h}{\sqrt{3}} \right)^2 (h) = \frac{\pi h^3}{9} \end{aligned}$$



$$\text{ii) } \frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt} \quad \text{and} \quad \frac{dV}{dh} = \frac{\pi h^2}{3}$$

$$\therefore \frac{dV}{dt} = \frac{\pi h^2}{3} \cdot \frac{dh}{dt}$$

$$\therefore 3\pi = \frac{\pi (81)}{3} \cdot \frac{dh}{dt} \rightarrow \frac{dh}{dt} = \frac{1}{9} \text{ cm} \cdot \text{sec}^{-1}$$

- c) Find a general solution to $\cos 2x = \sin x$

[2]

Solution:

$$\begin{aligned}\therefore 1 - 2\sin^2 x &= \sin x \\ \therefore 2\sin^2 x + \sin x - 1 &= 0 \\ \therefore (2\sin x - 1)(\sin x + 1) &= 0 \\ \therefore \sin x &= \frac{1}{2} \text{ or } \sin x = -1 \\ \therefore x &= n\pi + (-1)^n \left(\frac{\pi}{6}\right) \text{ or } \frac{\pi}{2}(4n+3)\end{aligned}$$

Note: This may also be answered in degrees.

Question 6:

- a) A particle is moving in a straight line with an initial velocity of 48ms^{-1} .

At any time t , the acceleration in ms^{-2} is given by $\ddot{x} = -6t$. Find

- i) The distance from its initial position when it stops
ii) The velocity as which it is travelling when it returns to its initial position.

[2]

Solution:

$$\begin{aligned}\ddot{x} &= -6t \\ \dot{x} &= -3t^2 + c \\ \text{at } t = 0, \dot{x} &= 48 = c \\ \therefore \dot{x} &= -3t^2 + 48 \\ x &= -t^3 + 48t + c_2 \\ \text{at } t = 0, x &= 0 = c_2 \\ \therefore x &= -t^3 + 48t \\ \text{at } x = 0, 3(-t^2 + 16) &= 0 \rightarrow t = 4 \\ \text{at } t = 4, x &= -4^3 + 48 \times 4 = 128 \quad \text{Distance is } 128\text{m} \\ \text{at } x = 0, -t^3 + 48t &= 0 \rightarrow t = \sqrt{48} = 4\sqrt{3} \\ \text{at } t = 4\sqrt{3}, v &= -3(4\sqrt{3})^2 + 48 = -96 \quad \text{Velocity is } -96\text{ms}^{-1}\end{aligned}$$

- b) A particle moving on a horizontal line has a velocity $v(\text{ms}^{-1})$, given by
 $v^2 = 64 + 24x - 4x^2$

- i) Prove the particle moves in simple harmonic motion

[1]

- ii) Find the centre of motion

[1]

- iii) Write down the period and amplitude

[2]

- iv) If initially, the particle is at the centre of motion and is moving to the left, write down an expression for the displacement as a function of time.

[1]

Solution:

i) $\frac{1}{2}v^2 = 32 + 12x - 2x^2$

$\ddot{x} = \frac{d}{dx}\left(\frac{v^2}{2}\right) = 12 - 4x = -4(x - 3)$ which is the form of SHM

- ii) At $\ddot{x} = 0, x = 3$

iii) $n^2 = 4 \rightarrow n = 2, \therefore T = \frac{2\pi}{2} = \pi$

At $v = 0, 4x^2 - 24x - 64 = 0 \rightarrow 4(x - 8)(x + 2) = 0$

Particle moves from -2 to 8, so amplitude is 5

iv) $x = 3 - 5\sin 2t$

- c) i) Write the expression $\sqrt{3}\cos\theta + \sin\theta$ in the form $R\cos(\theta - \alpha)$

[1]

- ii) Hence or otherwise solve $\sqrt{3}\cos\theta + \sin\theta = 1$, for $0^\circ \leq \theta \leq 360^\circ$.

[2]

Solution:

i) $R\cos(\theta - \alpha) = R(\cos\theta\cos\alpha + \sin\theta\sin\alpha)$

$\sqrt{3}\cos\theta + \sin\theta = 2\left(\frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta\right), \therefore R = 2, \theta = \frac{\pi}{6}$

$\therefore R\cos(\theta - \alpha) = 2\cos\left(\theta - \frac{\pi}{6}\right)$

ii) $2\cos\left(\theta - \frac{\pi}{6}\right) = 1 \rightarrow \cos\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2}$

$\therefore \theta - \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3} \rightarrow \theta = \frac{\pi}{2}, \frac{11\pi}{6}$

Question 7:

- a) The maximum charge of a battery that requires recharging is 12 volts. After 1 minute on the recharger, the charge is measured as 6 volts and after a further minute the charge has risen to 8 volts. Assuming the rate of increase of the charge is given by $\frac{dQ}{dt} = k(12 - Q)$,

i) Show this equivalent to $Q = 12 - Ae^{-kt}$

[2]

- ii) Find the value of the charge on the battery at the time it is placed on the recharger.

[2]

- iii) Find the exact value of the battery after it has been on the recharger for 3 minutes.

[3]

Solution:

i) $Q = 12 - Ae^{-kt}$

$$\frac{dQ}{dt} = k(12 - Q)$$

and since $Ae^{-kt} = 12 - Q$, $\frac{dQ}{dt} = k(12 - Q)$

ii) $Q = 12 - Ae^{-kt}$

at $t = 1, Q = 6 \rightarrow \therefore 6 = 12 - Ae^{-k}$ and $Ae^{-k} = 6$ or $A = 6e^k$

at $t = 2, Q = 8 \rightarrow \therefore 8 = 12 - Ae^{-2k}$ and $Ae^{-2k} = 4$ or $A = 4e^{2k}$

Equating for A , $6e^k = 4e^{2k} \rightarrow \frac{3}{2} = 3^k$

Since $A = 6e^k$, $A = 6(\frac{3}{2}) = 9$ and $Q = 12 - 9e^{-kt}$

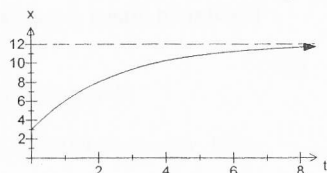
At $t = 0, Q = 12 - 9e^0 = 3$

iii) At $t = 3, Q = 12 - 9e^{-3k}$

$$= 12 - 9(e^{-k})^3$$

Since $e^k = \frac{3}{2}, e^{-k} = \frac{2}{3}$

and $Q = 12 - 9(\frac{2}{3})^3 = 9\frac{1}{3}$



- b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. If PQ passes through the point $(4a, 0)$

i) Show that $pq = 2(p + q)$

[2]

ii) Find the locus of M, the midpoint of PQ.

[3]

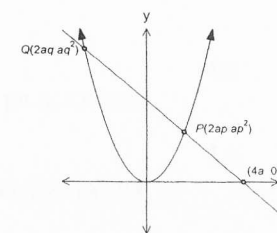
Solution:

i) $m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq} = \frac{a(p - q)(p + q)}{2a(p - q)} = \frac{p + q}{2}$

$\therefore$ Eqn PQ is $y - ap^2 = \frac{p + q}{2}(x - 2ap)$

$$y - ap^2 = \left(\frac{p + q}{2}\right)x - ap^2 - apq \rightarrow y = \left(\frac{p + q}{2}\right)x - apq$$

at $(4a, 0)$, $0 = \left(\frac{p + q}{2}\right)4a - apq \rightarrow pq = 2(p + q)$



ii) Midpoint PQ is $a(p + q), \frac{a}{2}(p^2 + q^2)$

Let $x = a(p + q)$, $y = \frac{a}{2}(p^2 + q^2)$

$$\frac{x}{a} = p + q \rightarrow \frac{2x}{a} = 2(p + q), \therefore \frac{2x}{a} = pq$$

Write y as $y = \frac{a}{2}\left\{(p + q)^2 - 2pq\right\}$

$$\therefore y = \frac{a}{2}\left\{\left(\frac{x}{a}\right)^2 - 2\left(\frac{2x}{a}\right)\right\}$$

$$= \frac{a}{2}\left\{\frac{x^2}{a^2} - \frac{4x}{a}\right\}$$

$$= \frac{x^2}{2a} - 2x$$

$$\therefore 2ay = x^2 - 4ax$$

$$\text{or } 2ay + 4a^2 = x^2 - 4ax + 4a^2 \rightarrow (x - 2a)^2 = 2a(y + 2a)$$